

DIHWorld Audio

Color / Amp / Tape Technical Manual

Current product versions, controls, presets, meters, and placement

DIHWorld Audio builds connected production systems around the Imhotep 369 Engine—a Triadic Harmonic Lattice—and Ma'at 357 / MomentumBend. Phi-derived control and the canonical Phi/2 ceiling preserve movement, balance, and musical intent without brickwall limiting.

This edition documents 4 current products. Release numbers are taken from the installed DIHWorld Audio Audio Units verified on 2026-10-05; they are product versions, not a shared catalog version.

Canon reference: normalized $\text{Phi}/2 = 0.809016994$. MomentumBend shapes motion and containment; the ceiling is a canonical boundary, not permission for hard limiting.

Operating law: begin from reset or a purpose-matched preset, compare at equal loudness, change one system at a time, and judge the result in the full arrangement.

APIS

RELEASE	FAMILY	PLACEMENT
v1.0.1	Color / Amp / Tape	Insert Safe

Body and color processor built around color, body, petal dispersion, air, protected I/O, and channel-mode support.

Recommended use: amp/color source tone

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
INPUT	Input	Float
COLOR	Color	Float
BODY	Body	Float
FLOWER	Flower Petal	Float
AIR	Air	Float
OUTPUT	Output	Float
channelmode	Channel Mode	Float
dualmonolink	Dual Mono Link	Float
gainmatch	Gain Match	Bool

Factory starting points — source-derived purpose

APIS CANON RESET
APIS CANON RESET — Sets the APIS body/air/flower balance for a deliberate harmonic field.

APIS BODY THRONE
APIS BODY THRONE — Centers Body and Color for a strong, stable source throne.

APIS AIR CROWN
APIS AIR CROWN — Raises Air and Flower Petal toward an open upper crown.

APIS BLOOM FIELD
APIS BLOOM FIELD — Spreads the flower/field relationship for a larger harmonic image.

APIS SHADOW CUT
APIS SHADOW CUT — Uses a restrained cut to clear space while leaving the source body readable.

APIS DUAL COMB
APIS DUAL COMB — Sets the APIS body/air/flower balance for a deliberate harmonic field.

APIS FULL FIELD
APIS FULL FIELD — Spreads the flower/field relationship for a larger harmonic image.

PTAHTUBE

RELEASE	FAMILY	PLACEMENT
v1.0.1	Color / Amp / Tape	Insert Safe

Phi-based tube color processor with input, drive, energy, balance, focus, output, and Flower of Life dispersal tied to its illuminated tube field.

Recommended use: tube color; vocal clarity required

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
input	Input	Float
drive	Drive	Float
output	Output	Float
energy	Energy	Float
balance	Balance	Float
power	Power	Float
focus	Focus	Float
flower	Flower of Life Dispersal	Bool
gainmatch	Gain Match	Bool
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Factory starting points — source-derived purpose

Canon Clear

Canon Clear — Returns the tube stage to a clean reference with low Drive and stable Balance.

Warm Filament

Warm Filament — Adds gentle filament warmth without asking the output control to create the effect.

Memphis Grit

Memphis Grit — Raises Drive/Energy for a rougher Memphis-style edge while Focus limits the mess.

Vocal Copper Edge

Vocal Copper Edge — Puts tube edge around a vocal center for articulation and body.

Bus Tube Glue

Bus Tube Glue — Uses moderate Drive and Balance for bus glue rather than a single-source distortion.

Drum Ember Forge

Drum Ember Forge — Forges drum transients with controlled tube energy and a firm center.

Ra Saturation

Ra Saturation — Sets a named tube/filament color with Drive, Energy, Balance, and Focus in a controlled relationship.

Khepri

RELEASE	FAMILY	PLACEMENT
v1.0.1	Color / Amp / Tape	Insert Safe

Sacred tape machine with selectable machine, Phi formula, tape type (Gold Alloy, Ferric Oxide, Metal Alloy), speed, drive, bias, wow, flutter, head bump, HF rolloff, and parallel mix.

Recommended use: tape/source color; default mix canon 1/Phi; secondary silver mastering-support dot

Customer formats: AU, VST3, Standalone

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
machine	Machine	Choice
formula	Phi Formula	Choice
tapeType	Tape Type	Choice
speed	Speed	Choice
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
input	Input	Float
drive	Drive	Float
bias	Bias	Float
wow	Wow	Float
flutter	Flutter	Float
bump	Head Bump	Float
rolloff	HF Rolloff	Float
mix	Mix	Float
output	Output	Float

Factory starting points — source-derived purpose

KHEPRI CANON RESET

KHEPRI CANON RESET — Keeps the tape machine clean and centered for a print/reference pass.

KHEPRI CLEAN PRINT

KHEPRI CLEAN PRINT — Keeps the tape machine clean and centered for a print/reference pass.

KHEPRI VOCAL SILK

KHEPRI VOCAL SILK — Uses tape speed, bias, and drive to smooth vocal edges without losing diction.

KHEPRI BASS FOUNDATION

KHEPRI BASS FOUNDATION — Adds tape foundation and low-frequency cohesion to a bass source.

KHEPRI DRUM PUNCH

KHEPRI DRUM PUNCH — Adds punch and tape cohesion to the drum bus while retaining the transient.

KHEPRI BUS GLUE

KHEPRI BUS GLUE — Uses moderate machine Drive and Bias for glue on a group bus.

KHEPRI AIR REPRO

KHEPRI AIR REPRO — Keeps the tape path open enough for air reproduction rather than darkening the source.

KHEPRI VINTAGE DRIFT

KHEPRI VINTAGE DRIFT — Introduces controlled speed/wow/bias movement for audible vintage drift.

KHEPRI DUAL REEL

KHEPRI DUAL REEL — Introduces controlled speed/wow/bias movement for audible vintage drift.

Chain: Pre-PtahTube

Chain: Pre-PtahTube — Marks KHEPRI's intended position in a DIH chain; load it as a routing stage, not a standalone color guess.

Chain: Vocal Body

Chain: Vocal Body — Uses tape speed, bias, and drive to smooth vocal edges without losing diction.

Chain: Drum Bus Pre-Glue

Chain: Drum Bus Pre-Glue — Adds punch and tape cohesion to the drum bus while retaining the transient.

Chain: Music Bus Foundation

Chain: Music Bus Foundation — Uses moderate machine Drive and Bias for glue on a group bus.

Chain: Clean Print Finish

Chain: Clean Print Finish — Keeps the tape machine clean and centered for a print/reference pass.

Meters and visualizers

INPUT VU: Ballistic input level. Calibrated to the printed analog VU scale.

OUTPUT VU: Ballistic post-tape level. Compare with INPUT to avoid loudness bias.

Waveform: Recent live output samples. Shows real signal motion, not a decorative animation.

Reels / beetle glow: Tape motion and drive state. Operational feedback; not a calibrated level meter.

SOVEREIGN

RELEASE	FAMILY	PLACEMENT
v1.0.1	Color / Amp / Tape	Mix Bus

Source-oriented amplifier and tone stage with drive, bass, mid, treble, presence, and master controls.

Recommended use: stronger color and bus authority; secondary silver mastering-support dot

Customer formats: AU, VST3

Exposed parameter surface

Parameter ID	Display name	Type
power	Power	Bool
drive	Drive	Float
bass	Bass	Float
mid	Mid	Float
treble	Treble	Float
presence	Presence	Float
master	Master	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
gainmatch	Gain Match	Bool

Factory starting points — source-derived purpose

ROYAL BALANCE [Stereo]

ROYAL BALANCE — Sets Sovereign’s drive/EQ authority for the named source role. Stereo variant: keep the linked center intact and audition width in the full mix.

CLEAN CROWN [Stereo]

CLEAN CROWN — Sets Sovereign’s drive/EQ authority for the named source role. Stereo variant: keep the linked center intact and audition width in the full mix.

TAPE FRONT [Stereo]

TAPE FRONT — Uses Drive and tone lanes for a tape-front color before the final Master trim. Stereo variant: keep the linked center intact and audition width in the full mix.

GUITAR ROYALTY [Stereo]

GUITAR ROYALTY — Shapes guitar body, mid authority, and presence with a finished royal contour. Stereo variant: keep the linked center intact and audition width in the full mix.

BASS BODY AMP [Stereo]

BASS BODY AMP — Adds bass-body authority and controlled drive without replacing the sub foundation. Stereo variant: keep the linked center intact and audition width in the full mix.

VOCAL AUTHORITY [Stereo]

VOCAL AUTHORITY — Gives a vocal a forward authority contour through Mid, Presence, and Master. Stereo variant: keep the linked center intact and audition width in the full mix.

PARALLEL COLOR BUS [Stereo]

PARALLEL COLOR BUS — Sets a parallel color bus so the dry source can retain its transient identity. Stereo variant: keep the linked center intact and audition width in the full mix.

DARK IRON CAB [Stereo]

DARK IRON CAB — Moves the source toward a dark iron cabinet contour with less upper sheen. Stereo variant: keep the linked center intact and audition width in the full mix.

ROYAL BALANCE [Mono]

ROYAL BALANCE — Sets Sovereign’s drive/EQ authority for the named source role. Mono variant: collapse the field and use it as a center-focused insert.

CLEAN CROWN [Mono]

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ROYAL BALANCE [Dual Mono]

ROYAL BALANCE — Sets Sovereign's drive/EQ authority for the named source role. Dual-mono variant: preserve independent left/right decisions for asymmetric sources.

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